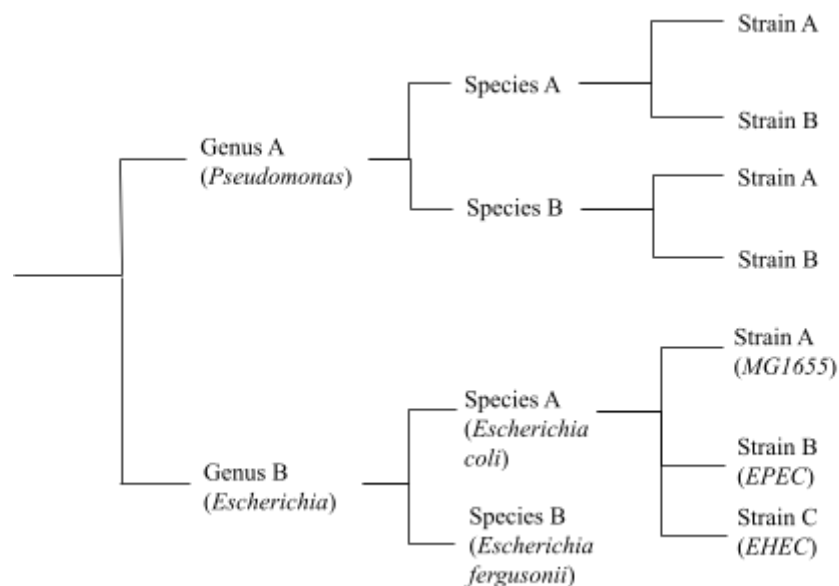


## Transfer Learning At The Strain and Genus Levels

### Framework

Humans have long been attempting to elucidate the evolutionary relationships between organisms across all levels of biological classification, including species (more specifically different strains) and genera. One way to achieve this is through the use of phylogenetic trees in which branches are used to track the relationships between organisms (that is to say, which organisms are ancestors of which). According to this organizational technique, strains have shorter branches connecting them to their most recent common ancestor, followed by species and thus genera would have the greatest phylogenetic distance between them.



Previously, this tracking was performed by sequence alignment algorithms comparing DNA, RNA or protein sequences. However, with the rise in computer science and biotechnology, researchers have increasingly been able to investigate questions related to phylogenetic inference with various machine learning techniques.

Machine learning is a sector of artificial intelligence centered around pattern recognition. It involves using one dataset to train algorithms to identify patterns and thus make accurate inferences about a different dataset (Bergmann 2025). An example of machine learning is transfer learning which involves two training phases and an assessment phase. The second training phase is used to fine-tune the algorithm in an attempt to optimize performance before assessment. This can be achieved by training the algorithm on two different sets of data which broadens the encoder's ability to identify patterns in the assessment data.

One technique that can utilize transfer learning is a neural network which is a clustering tool that works by applying filters to the training dataset. The input data is provided to layers of neurons (also known as nodes) which turns on an activation function. The network is then trained

on this input data by adjusting the edge weights between adjacent nodes after evaluating the quality of the output (Tan et al. 2017).

An example of a neural network is an autoencoder which can be trained to reproduce the input data. This is a type of unsupervised learning meaning the data that is used to train the model is unlabeled, placing the focus on identifying patterns within it (Mo, Hahn, and Smith 2024). A denoising autoencoder (DAE) is a specific kind of autoencoder in which the input is obscured with added noise.

Despite neural networks proficiency in discerning information from complex datasets, it remains largely untested on model microbes such as *Escherichia coli* (*E.coli*) and *Pseudomonas aeruginosa* (*P. aeruginosa*) despite both having several compendia of their respective gene expression data available publicly. This is partially due to the extent of the performance of machine learning models is unclear, partly due to strain variation. Strain variation refers to changes in genes at the sub-species level which, despite being an important component of understanding the phenotypic variation within a species, inhibits the applicability of gene expression data to an entire species (Lee et al. 2022). Two prevalent disease-causing strains of *E. coli* are *Enteropathogenic E. coli* (EPEC) and *Enterohemorrhagic E. coli* (EHEC) which have increased virulence. These strains are known for their antibiotic-resistance and are also significant causes of mortality, especially in developing countries with less reliable sanitation systems.

## Goals

We intend to answer the question: does strain-level transfer learning yield results that are more concordant with biological databases, such as EcoCyc or [pseudomonas.com](https://pseudomonas.com), than species-level and genus-level transfer learning? We will test the accuracy of genus-level transfer learning in DAEs as research at this level of classification has yet to be performed and we already have some unpublished preliminary studies at the species level that suggest its existence (Doing). Of the three levels that we will test, strains are the most genetically similar to each other, followed by species and then genera. Consequently, in our testing of strain-level distances, we hypothesize that transfer learning will produce the most accurate results at the strain-level, then the species level, and the least accurate results at the genus-level. We anticipate that our autoencoder will be able to identify the most statistically significant patterns in the genomes of different strains of the same species. We expect that the DAE-identified patterns will be slightly less accurate when comparing genomes of different, but still closely related, species (in this case, *E. coli* and *P. aeruginosa*) and the least accurate across different genera.

## Impact

If our findings suggest that autoencoders can identify statistically-significant patterns across strains, they could serve as evidence that we could bridge the gap between laboratory-studied *E. coli* strains and disease-causing strains. Relations between the EPEC strain and the EHEC strain could be especially important due to their increased virulence. This would

also significantly improve our understanding of this bacteria and allow us to develop a more efficient public health response.

Additionally, providing some evidence that transfer across genera is possible could further advance the field of computational biology. It would create a basis for further investigations to test the extent to which machine learning can be transferred and could solidify existing studies conducted across genera. One area that would be especially impacted includes studies conducted on laboratory animals, such as mice, in which the researchers have attempted to generalize the results to humans. It would also provide a basis for extending existing findings to understudied organisms. This would allow us to gain a deeper understanding of a variety of organisms that are difficult to study in a laboratory setting for a variety of reasons such as limited resources.

This research will also advance the field of computational biology towards being able to ask other questions regarding phylogenetic inference. We currently only have data sets for certain organisms, but with further investigation, this research could be extended to a larger sample and greater variety of organisms. This could allow us to determine the maximum possible phylogenetic branch length between microbes at which our denoising autoencoder can recognize biologically-driven gene expression patterns.

## **Plan**

For each condition that we test, there will be two sections of the experiment: the first section involves the experimental portions (a data setup phase and a transfer phase) and the second section consists of analyzing the output of the autoencoder. We have a compendium of approximately twenty thousand samples of *E. coli* with strain-level labels that already existed and consists of all publicly available data as of 2025. We also have a compendium of approximately two thousand samples of *P. aeruginosa* that already existed and consists of all publicly available data as of 2023 (Doing, Georgia et al. 2023). Samples in both compendia are from a variety of conditions, including, but not limited to, different studies, temperatures, media and strains. In the data setup phase, we will use the *E.coli* and *P. aeruginosa* compendia to develop criteria to determine which samples will be used for which step of the transfer phase.

The transfer phase will include three parts: pre-training, fine tuning and assessment. A different set of data samples will be used in each phase depending on which experimental condition we are testing at that time. Some conditions include comparing strains of *E. coli* by pre-training on the MG1655 strain and fine tuning it on other strains such as *Enterotoxigenic E. coli* and DH10B, comparing the EHEC and EPEC strains, and pre-training on *E. coli* samples and fine tuning using *P. aeruginosa* samples. Our model will be constructed using the Keras Python library.

Given that a DAE is a clustering technique, our analysis will consist of looking for enrichment between the output and known pathways on pseudomonas.com and KEGG pathways (well-known gene pathway databases). This mapping between each gene and pathway will be used to evaluate the biological relevance of the gene signatures (Tan, Jie, et al. 2017). We will

also compare outputs across different pairs of pre-training and fine-tuning datasets. For example, we will compare the results from our *E. coli* strain condition to those of our *E. coli* and *P. aeruginosa* test to determine the strain-level workflow and the genus-level workflow of our DAE.

### **Personal Goals**

As a neuroscience major, one of the areas that I am deeply interested in is the overlap between biology and computer science. Thus this research is the perfect opportunity for me to expand my own knowledge while simultaneously contributing to the discipline and society as a whole. I also hope to one day work in the field of bioinformatics and this opportunity would serve as an incredible introduction to the technical aspects of the field which are often more difficult to learn in a lecture-based classroom setting.

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